Tracing the Longest Common Subsequence (LCS)

Problem

Given two strings:

- X = ABCBDAB
- Y = BDCAB

Find the LCS (Longest Common Subsequence). Two valid results are: **BDAB** and **BCAB**.

DP Table with LCS Lengths

	-	B	D	C	A	B
-	0	0	0	0	0	0
A	0	0	0	0	1	1
B	0	1	1	1	1	2
C	0	1	1	2	2	2
B	0	1	1	2	2	3
D	0	1	2	2	2	3
A	0	1	2	2	3	3
B	0	1	2	2	3	**4**

Traceback for LCS = `BDAB`

- Start at cell (7,5): B == B $\Rightarrow$ move diagonally (6,4) $\rightarrow$ add B
- At (6,4): A == A $\Rightarrow$ move diagonally (5,3) $\rightarrow$ add A
- At (5,3): B != C $\Rightarrow$ max(dp[4][3], dp[5][2]) $\rightarrow$ move up to (4,3)
- At (4,3): B != C $\Rightarrow$ move up to (3,3)
- At (3,3): C == C $\Rightarrow$ move diagonally (2,2) $\rightarrow$ add C (skip this for BDAB, continue)
- Backtrack to (2,2): B != D $\Rightarrow$ move left to (2,1)
- At (2,1): B == B $\Rightarrow$ move diagonally (1,0) $\rightarrow$ add B

  Result: Reverse(B, A, D, B) = BDAB

Traceback for LCS = `BCAB`

- Start at cell (7,5): B == B $\Rightarrow$ move diagonally (6,4) $\rightarrow$ add B
- At (6,4): A == A $\Rightarrow$ move diagonally (5,3) $\rightarrow$ add A
- At (5,3): D != C $\Rightarrow$ move up to (4,3)
- At (4,3): B != C $\Rightarrow$ move up to (3,3)
- At (3,3): C == C $\Rightarrow$ move diagonally (2,2) $\rightarrow$ add C

- At (2,2): B != D $\Rightarrow$ move left to (2,1)
- At (2,1): B == B $\Rightarrow$ move diagonally (1,0) $\rightarrow$ add B

Result: Reverse(B, A, C, B) = BCAB

Conclusion

Both BDAB and BCAB are valid LCS solutions of length 4. The DP table allows us to backtrack along multiple valid paths, depending on whether we choose to move left or up when there's a tie.